EXTENDED EQUAL AREA CRITERION
JUSTIFICATIONS, GENERALIZATIONS, APPLICATIONS

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Abstract. The extended equal area criterion (EEAC) is revisited, with the following three objectives. First, to systematically state the main hypotheses and key conditions underlying it, to justify the former and suggest means to guarantee the latter. Critical machines' identification and error analysis are among the investigated issues. Second, to scan all possible types of instability likely to arise in practice and to devise means to treat them. The extension of EEAC to cases beyond the so-called "first swing" stability makes it more robust than all direct methods developed up to now. The third main objective is to extract essential information out of a large body of simulations, and to show that the above improvements and extensions enhance indeed the EEAC accuracy and its capability to work properly even under stringent conditions. Possible EEAC applications are also discussed, and uses of the method as such or as an auxiliary technique for more sophisticated approaches are suggested.

Keywords : on-line transient stability assessment, analytic stability analysis, second swing instability.

1. INTRODUCTION

Since the beginning of its development [1,2], the *extended equal area criterion* (EEAC) has shown very interesting possibilities for on-line *transient stability assessment* (TSA). A significant advantage is the algebraic expressions it provides for the calculation of critical clearing times and stability margins. This makes the conventional TSA particularly easy, and furnishes analytic sensitivity tools and therefrom means to control.

The research work on EEAC has pursued two complementary directions : one concerns enhancement of accuracy and robustness. The other deals with sensitivity analysis aspects [3].

This paper concentrates on the former direction. It comprises three main parts, corresponding to the three objectives specified in the abstract.

The first part is composed of Sections 2,3, and 4. Section 2 restates the essentials of the EEAC, summarizes the basic formulation, then scrutinizes an important assumption underlying it and proposes means of justification. Sections 3 and 4 deal with accuracy aspects of the method. Section 3 discusses the critical machine (s)' identification; means to draw up an appropriate list of candidate critical clusters are suggested, and the multimachine critical cluster is modelled. Section 4 explores the possible sources of errors, decomposes them into two types, then proposes means to circumvent or reduce them to the extent possible.

The second part, consisting of Section 5, investigates and brings out an important generalization of the EEAC : that of treating any type of instability, i.e. "first swing" as well as "second swing" instability. Note that this is the first time that a direct method succeeds in tackling stability beyond the first swing. Indeed, if from a theoretical viewpoint the Liapunov criterion should be able to depict (although "blindly") such second swing instabilities, the way this criterion is applied in practice prevents it from such a possibility. The extension proposed in this paper makes second swing phenomena extremely easy to identify and treat.

The third part of the paper (Section 6) reports essential information extracted out of over two thousand exhaustive simulation results, performed on 10 different power systems, under various operating conditions. The accent is put on the cases which do not work satisfactorily enough, in order to get further in-depth understanding.

2. THE EXTENDED EQUAL AREA CRITERION

2.1. Principle

For an assigned disturbance, the multimachine system is decomposed into two subsets : one consisting of the "cluster of critical machine(s)" to which we henceforth refer to as the "critical cluster", the other comprising the remaining system. The two subsets are further transformed into two equivalent machines; each one of them models the corresponding machines' dynamics within a *partial center of angles* (PCOA) frame. Finally, the equivalent two-machine system is reduced into a "*one-machine-infinite-bus*" (OMIB) system; the application of the equal area criterion to this latter provides means to the multimachine system TSA.

Two different TSA measures are derived : the *critical clearing time* (CCT) and the transient stability margin. They both take on simple, extremely easy to use, algebraic expressions, thanks to the conjunction of the OMIB system with a modified Taylor series expansion.

Remark. The success of EEAC strongly depends upon:
- the correct identification and modelling of the critical cluster (see § 2.4 and Section 3);
- the soundness of the remaining system's replacement by the OMIB system (see § 2.5);
- the way of circumventing the errors introduced by the OMIB and by the truncation of the Taylor expansion (see Section 4).

2.2. Basic formulation

2.2.1. *Multimachine system model.* With the standard simplifying assumptions, the motion of an n-machine system is described by

$$\dot{\delta}_i = \omega_i \quad , \quad M_i\dot{\omega}_i = P_{mi} - P_{ei} \qquad i = 1,2,\dots,n \tag{2.1}$$

where

$$P_{ei} = E_i^2 Y_{ii} \cos\theta_{ii} + \sum_{j=1,j\neq i}^{n} E_i E_j Y_{ij} \cos(\delta_i - \delta_j - \theta_{ij}) \tag{2.2}$$

and where the following standard notation is used :

δ_i (ω_i)	rotor angle (speed)
M_i	inertia coefficient
P_{mi} (P_{ei})	mechanical input (electrical output) power
E_i	voltage behind the direct axis transient reactance
$\underset{\sim}{Y}$	admittance matrix reduced at the internal generator nodes
Y_{ij} (θ_{ij})	modulus (argument) of the ij-th element of $\underset{\sim}{Y}$.

M_i, P_{mi} and E_i are assumed to be constant throughout the transients; all loads are modelled as constant impedances.

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2.2.2. Equivalent two-machine aggregated model. Denoting by

s the specific critical cluster assumed momentarily to consist of one machine only
A the set of all remaining machines
a its equivalent, aggregated to one, machine

we set

$$M_a = \sum_{\ell \in A} M_\ell \; ; \; \delta_a = M_a^{-1} \sum_{\ell \in A} M_\ell \delta_\ell \; ; \; \omega_a = \dot{\delta}_a \; ; \; \gamma_a = \ddot{\delta}_a \qquad (2.3)$$

and get the motion of the PCOA of the set A

$$M_a \ddot{\delta}_a = \sum_{\ell \in A} (P_{m\ell} - P_{e\ell}) \qquad (2.4)$$

Upon assuming

$$\delta_\ell = \delta_a \quad \forall \ell \in A \qquad (2.5)$$

we have

$$P_{e\ell} = E_\ell^2 Y_{\ell\ell} \cos\theta_{\ell\ell} + E_\ell E_s Y_{\ell s} \cos(\delta_a - \delta_s - \theta_{\ell s}) + + \sum_{j \in A, j \neq \ell} E_\ell E_j Y_{\ell j} \cos\theta_{\ell j} \qquad (2.6)$$

For the specific candidate critical machine, we get

$$M_s \ddot{\delta}_s = P_{ms} - P_{es} \qquad (2.7)$$

$$P_{es} = E_s^2 Y_{ss} \cos\theta_{ss} + \sum_{\ell \in A} E_s E_\ell Y_{s\ell} \cos(\delta_s - \delta_a - \theta_{s\ell}) \; . \qquad (2.8)$$

2.2.3. Equivalent OMIB system. Setting

$$\delta \triangleq \delta_s - \delta_a \qquad (2.9)$$

we get

$$M\ddot{\delta} = P_m - [P_c + P_{max} \sin(\delta - \nu)] \quad . \qquad (2.10)$$

where

$$\left.\begin{aligned} M &= M_s M_a M_T^{-1} \quad , \quad M_T = \sum_{i=1}^{n} M_i \\ P_m &= (M_a P_{ms} - M_s \sum_{\ell \in A} P_{m\ell}) M_T^{-1} \\ P_c &= (M_a E_s^2 G_{ss} - M_s \sum_{\ell, j \in A} E_\ell E_j G_{\ell j}) M_T^{-1} \\ P_{max} &= (C^2 + D^2)^{1/2} \quad ; \quad \nu = -\tan^{-1}(C/D) \\ C &= (M_a - M_s) M_T^{-1} \sum_{\ell \in A} E_s E_\ell G_{s\ell} \quad ; \quad D = \sum_{\ell \in A} E_s E_\ell B_{s\ell} \end{aligned}\right\} \qquad (2.11)$$

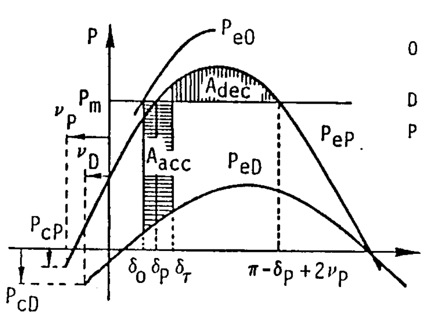
2.2.4. Stability measures. They derive from the well-known equal area criterion applied to (2.10). Fig.1 illustrates the plot of the P-δ curves provided in the pre-fault or original (O), during-fault (D), and post-fault (P) configurations. The original (steady-state) operation is characterized by the rotor angle δ_o located at the crossing of the horizontal line $P = P_m$ with the original P_{eO} curve, partially drawn. The post-fault stable (respectively unstable) equilibrium point is determined by the intersection of P_m with P_{eP} ; this provides δ_p (respectively $(\pi - \delta_p + 2\nu_p)$). Moreover, the value that the angle reaches at the fault clearing time delimits the accelerating area A_{acc} and decelerating area A_{dec} , which measure the corresponding transient energies :

$$A_{acc} = (P_m - P_{cD})(\delta_\tau - \delta_o) + P_{max\,D}[\cos(\delta_\tau - \nu_D) - \cos(\delta_o - \nu_D)] \qquad (2.12)$$

$$A_{dec} = (P_{cP} - P_m)(\pi - \delta_\tau - \delta_p + 2\nu_p) + P_{max\,P}[\cos(\delta_\tau - \nu_p) + \cos(\delta_p - \nu_p)] \; . \qquad (2.13)$$

The EEAC provides two alternative TSA measures : the transient stability margin

$$\eta = A_{dec} - A_{acc} \qquad (2.14)$$



O : *original (pre-fault)*
D : *during-fault*
P : *post-fault*

FIGURE 1 δ *(Taken from [2])*

relative to a given clearing time τ and its corresponding clearing angle δ_τ , and the conventional CCT corresponding to

$$\eta = 0 \; . \qquad (2.15)$$

2.3. CCT (t_c) computation

For a given disturbance and its corresponding critical cluster, this computation amounts to :
(i) computing δ_c , by solving (2.15) with $\delta_\tau = \delta_c$;
(ii) computing t_c by solving eq. (2.16) :

$$\delta_c = \delta_o + \alpha_1^{-1}\alpha_2^{-2}\gamma t_c^2/2 + \alpha_1^{-1}\alpha_2^{-4}\gamma^{(2)} t_c^4/24 \qquad (2.16)$$

where γ and $\gamma^{(2)}$ denote the acceleration and its second derivative of the OMIB system at $t = 0^+$, i.e. immediately after the disturbance inception :

$$\gamma = \gamma_s - \gamma_a = M_s^{-1}(P_{ms} - P_{es}) - M_a^{-1} \sum_{\ell \in A} (P_{m\ell} - P_{e\ell}) \; . \qquad (2.17)$$

Eq. (2.16) is a Taylor series truncated after the t^4 term, in which the corrective factors α_1 and α_2 have been introduced to counterbalance truncation effects [2].

The above computation is extremely fast and almost negligible with respect to that of the $\underline{Y}$ matrix reduction.

2.4. Identification of the critical cluster

For a given disturbance the procedure consists of :
(i) drawing up a list of "candidate" critical clusters, by using the generators' initial accelerations to "pre-filter" the plausible candidates;
(ii) computing by turn the corresponding candidate critical clearing times : the actual critical cluster will be the one yielding the smallest among the above values; this is considered to be the actual CCT provided by the EEAC.

Observe that this identification amounts to using repeatedly the procedure of § 2.3; note, however, that for a given disturbance, $\underline{Y}$ reduction is performed only once for all candidate clusters. The required computing effort is still almost negligible [2].

2.5. On the basic hypotheses underlying the EEAC

The EEAC relies on one conjecture and one assumption.

CONJECTURE. The disturbed system's separation depends upon the angular deviation between the following two - and only these two - equivalent clusters : the critical one, (s) , and that comprising the remaining machines, (a) . This conjecture amounts to declaring the loss of synchronism to be reached when the angular deviation $\delta = \delta_s - \delta_a$ reaches $(\pi - \delta_p + 2\nu_p)$. We believe that, irrespective of the method used, this criterion is more sound than the conventional one, according to which instability arises as soon as the largest rotor angular deviation among *any* two machines reaches about 180°. It is also more sound to use the PCOA than the "global COA" which incorporates the critical cluster.

The example given in the closure of [2] corroborates the above considerations.

ASSUMPTION. Within an aggregated cluster, the rotor angles of the individual machines are supposed to be represented by the corresponding PCOA, i.e.,

$$\delta_\ell = \delta_a \quad \forall \ell \in A \quad . \qquad (2.5)$$

To justify this assumption, let us evoke the following arguments.
(i) As aforementioned, throughout our simulations we have found that the key point of a system's separation is the motion of the two PCOA's. In other words, the transient energies responsible for the separation are those between the two PCOA's and not between any two individual machines.
(ii) It is clear that assuming $\delta_\ell = \delta_a$ $\forall \ell \in A$ throughout both the during-fault and post-fault phases would indeed have been unsound. This however is not what is actually implied. Indeed, the EEAC belongs to the class of direct methods : to assess stability, it uses information available at the very beginning of the post-fault phase; in other words, it needs knowledge of the system evolution

only up to the critical clearing time (and hence angle), along with the P_{max} values ($P_{max\,D}$, $P_{max\,P}$). For this short time interval, the assumption (2.5) is quite reasonable and physically sound.

(iii) The thousands of cases we have successfully run on ten different power systems with a wide variety of operating conditions validate the EEAC.

(iv) Let us consider an extreme situation which a priori seems to invalidate the assumption. This is the case of a three-phase short-circuit at bus #6, followed by tripping of line 6-11 in the 10-machine New England system described in [4].

According to the *step-by-step* (SBS) numerical integration, the CCT is found to be 0.215 s. According to EEAC, the following "candidate" CCT_i's are given (here i indicates the candidate critical generator(s), G_i): CCT_1 = 0.302 s; CCT_2 = 0.232 s; CCT_3 = 0.265 s; $CCT_{2,3}$ = 0.233 s. Hence, according to the EEAC, the CCT is 0.232 s and the critical generator is G_2; it should go unstable just before the cluster (G_2,G_3). Fig.2 illustrates this case for a clearing time just above 0.215 s. The curves are provided by an SBS program; they are drawn with respect to the appropriate *partial* COA; they are therefore slightly different from those of FiG. 4.9 of Ref.[4] where the *global* COA was considered. Notice that the angle $(\pi-\delta_p+2\nu_p)$ takes on the values 116°, 120°, and -142° for respectively the δ_2, $\delta_{2,3}$, and δ_1 cases. They predict that generators G_2, (G_2,G_3), and G_1 will successively lose synchronism. This is in perfect accordance with the predictions of the sole EEAC.

N.B. Apparently, the time evolution of δ_1 in Fig.2 is strongly against the assumption (2.5): without the justification given above in (ii), this assumption would look unfounded and the good behaviour of the EEAC surprising.

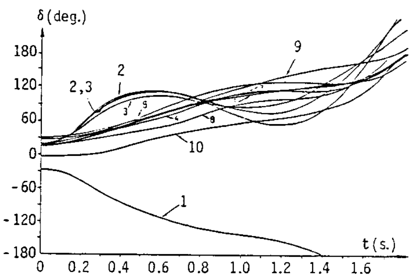


FIGURE 2

3. MULTIMACHINE CRITICAL CLUSTERS

3.1. General observations

As a rule, multimachine critical clusters generally arise for disturbances located at non-generator buses (at a center of a group of machines, or nearby weak tie-lines). All our test systems have exhibited existence of such clusters; some of their general features are summarized hereafter.

- Disregarding the multimachine critical clustering whenever it actually arises, may greatly deteriorate the EEAC accuracy in terms of CCT; the errors may be as large as 40 % in the case of disturbances located at generator buses and even larger in the non-generator bus cases.
- The composition of a critical cluster depends strongly upon the fault location, but is rather independent of the operating point; nevertheless, important topological changes may influence it. Therefore,
- the number of different critical clusters attached to a given bus is very limited, especially for clusters of more than 3 machines.

The identification of multimachine critical clusters is not trivial; some guidelines are discussed below.

3.2. On the critical cluster identification

Extensive simulations show that the procedure of § 2.4 based on the computation of candidate CCTs, is very efficient indeed for critical clusters composed of 1 or 2 machines. For multimachine clusters, however, where as many as half of the machines may actually swing against the others, this procedure becomes unefficient. This is because in such cases the critical cluster may comprise generators situated far from the disturbance location and whose individual candidate CCT may be rather large and hence may mask their possible contribution to the critical cluster.

Nevertheless, for a given power system, the general rules of § 3.1 allow the operator to preassign a list of candidate clusters attached to each interesting bus, on the basis of knowledge acquired off-line.

An interesting computational aspect is that the assessment of multimachine candidate critical clusters is as unexpensive as that for a single one. The CCT is thus unambiguously determined, provided that the actual critical cluster is included in the list; this is always possible at the expense of very little, on-line computational effort.

3.3. Critical cluster modelling

The general formulation of Section 2 readily yields the following formulae of the equivalent critical machine :

$$\left.\begin{aligned}
M_S &= \sum_{k\in S} M_k \quad ; \quad \delta_S = M_S^{-1} \sum_{k\in S} M_k\delta_k \\
P_m &= (M_a \sum_{k\in S} P_{mk} - M_S \sum_{\ell\in A} P_{m\ell})\, M_T^{-1} \\
P_c &= (M_a \sum_{k,j\in S} E_kE_jG_{kj} - M_S \sum_{\ell,j\in A} E_\ell E_j G_{\ell j})\, M_T^{-1} \\
C &= (M_a - M_S)\, M_T^{-1} \sum_{k\in S}\sum_{\ell\in A} E_kE_\ell G_{k\ell} \\
D &= \sum_{k\in S}\sum_{\ell\in A} E_kE_\ell B_{k\ell}
\end{aligned}\right\} \tag{3.1}$$

4. ACCURACY ANALYSIS

4.1. The two main types of errors

Observing that the EEAC amounts to computing the CCTs corresponding to candidate *critical clearing angles* (CCA or δ_c), we decompose the total error, ϵ, into two parts : (i) the error ϵ_1, introduced by the OMIB equivalent; (ii) the error ϵ_2, introduced by the Taylor expansion (2.16). Obviously, ϵ is a non-linear function of ϵ_1, ϵ_2 :

$$\epsilon = f(\epsilon_1,\epsilon_2) \quad . \tag{4.1}$$

To assess the above errors, we compare quantities of concern provided by the EEAC with those provided by the SBS. These quantities, identified respectively by the superscripts E and S, are represented in Fig.3; the curves represent the equivalent rotor angle evolution of the critical cluster with respect to the PCOA frame. We readily get :

$$\epsilon = t_c^E - t_c^S \quad ; \quad \epsilon_1 = \delta_c^E - \delta_c^S \quad ; \quad \epsilon_2 = t_c^E - t_1 \tag{4.2}$$

where (see Fig.3)

$$t_1 \overset{\Delta}{=} t\Big|_{\delta^S(t) = \delta_c^E} \quad . \tag{4.3}$$

4.2. Analysis of the ϵ_1-type errors

The main source of these errors is the assumption stated in § 2.5. Since δ_c is a non-linear function of δ_o, δ_p, δ_p^u (see Fig.1 and eq.(2.15)), we will concentrate on these latter, and on possible means of improving their accuracy.

(i) An easy way to avoid errors on δ_o is to take its load flow value, δ_o^{LF}, which anyhow is available at the beginning of the disturbed phase, rather than δ_o^E provided by the OMIB system.

(ii) Concerning δ_p, we suggest to use the corrective factor

$$\Delta\delta_o = \delta_o^{LF} - \delta_o^E$$

and compute accordingly (see Fig.1) :

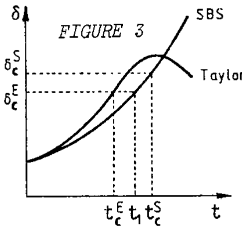


FIGURE 3

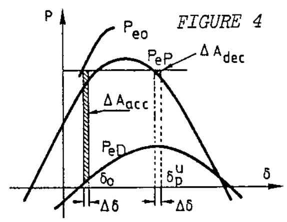


FIGURE 4

$$\delta_p' = \delta_p + \Delta\delta_o = \sin^{-1}\frac{P_m - P_{cP}}{P_{max\,P}} + \delta_o^{LF} - \sin^{-1}\frac{P_m - P_{cO}}{P_{max\,O}} \tag{4.4}$$

(iii) As for the error on δ_p^u , let us observe that according to Fig.4 the overall accuracy depends much more on that of δ_o than of δ_p^u (a same $\Delta\delta$ introduces a much larger discrepancy ΔA_{acc} than ΔA_{dec}). Hence, the corrected value

$$(\delta_p^u)' = \pi - \delta_p' + 2\nu_p \tag{4.5}$$

will generally provide good accuracy.

4.3. Analysis of the ϵ_2 -type errors

Use, in conjunction with the OMIB system, of a Taylor series truncated after the t^4 term, totally frees the EEAC from numerical integrations; as a counterpart, we have to face the ϵ_2 -type errors.

The Taylor series, expanded up to the t^4 term about the origin takes on the general form :

$$\delta(t) = \delta(0) + \delta^{(1)}t + \frac{1}{2}\delta^{(2)}t^2 + \frac{1}{6}\delta^{(3)}t^3 + \frac{1}{24}\delta^{(4)}t^4 \tag{4.6}$$

where $\delta^{(i)}$ denotes the i-th order derivative of δ evaluated at 0^+ . This expression may be used to treat a sequence of disturbances. In the particular case of a single disturbance, $\delta^{(2i+1)}(0) = 0 \;\; \forall i = 0,1,\dots$, and eq.(4.6) simplifies to

$$\delta(t) = \delta_o + \frac{1}{2}\gamma t^2 + \frac{1}{24}\gamma^{(2)}t^4 . \tag{4.7}$$

In general, truncating terms higher than t^4 introduces negligible errors when the clearing time t is small ($t < 0.3\,s$) ; these errors may however become important for large t's, depending upon the values of the neglected terms in $\gamma^{(4)}, \gamma^{(6)}$, etc ; this is often the case for disturbances located at non-generator buses.

To reduce the above errors, Refs.[1,2] propose the use of two corrective factors α_1, α_2 , with $1 > \alpha_1 > 0$, $\alpha_2 > 1$, and $\alpha_1\alpha_2^2 \simeq 1$. This procedure amounts to : 1° artificially shrinking the time interval of this series application by multiplying $[\delta(t) - \delta_o]$ by α_1 ; 2° computing the corresponding $t_\alpha < t$; 3° expanding t_α by setting $t = \alpha_2 t_\alpha$ to find the t corresponding to the $\delta(t)$ value of concern.

Let us explore further the optimal choice of α_1 , upon fixing

$$\alpha_1\alpha_2^2 = 1 . \tag{4.8}$$

The above procedure applied to (4.7) with $\delta = \delta_c$ yields

$$\delta_c = \delta_o + \frac{1}{2}\gamma t_c^2 + \frac{\alpha_1}{24}\gamma^{(2)}t_c^4 . \tag{4.9}$$

To make our reasoning more concrete, let us consider the real case consisting of a three-phase short-circuit applied at the non-generator bus #88 of the Greek system, for which the EEAC exhibits the largest discrepancy with respect to SBS ($t_c^S = 0.900$, $t_c^E = 0.732\,s$). For this case, the actual critical machine is found to be the generator G89. The OMIB rotor angle corresponds to $\delta = \delta_{89} - \delta_a$, i.e. to the time evolution of δ_{89} with respect to the PCOA frame. Fig.5 represents on one hand the curve $\delta^S(t)$ computed by SBS, on the other hand the curves $\delta^E(t)$ drawn by the Taylor expansion (4.8), for various values of α_1 . Many worthwhile observations may be made.

(i) The curves $\delta^E(t)$ drawn in the cases $\alpha_1 = 0$, $\alpha_1 = 1$ correspond respectively to totally neglecting $\gamma^{(2)}$ and fully taking it into account in (4.8). Observe that these extreme cases are justifiable respectively when $|\gamma|$ is far larger than $|\gamma^{(2)}|$, $|\gamma^{(4)}|$, etc. , and when $\gamma^{(4)}$ is negligible. Observe also that since the sign of $\gamma^{(4)}$ is almost always opposite to that of $\gamma^{(2)}$, neglecting partly $\gamma^{(2)}$ (by setting $\alpha_1 \in (0,1)$) amounts to compensating the truncated terms.

(ii) For those cases where $\gamma^{(2)}$ is actually small, as it mostly happens for faults located at important generator buses (where $\gamma^{(2)}, \gamma^{(4)}, \dots$ almost vanish), the $\delta^E(t)$ curves drawn with $\alpha_1 = 0$ and $\alpha_1 = 1$ almost coincide with each other and hence with $\delta^S(t)$: the accuracy of $\delta^E(t)$ becomes very good and almost independent of α_1 ; conversely, for large $\gamma^{(2)}$ values the accuracy will be very sensitive to α_1 .

(iii) Both of the curves $\delta^S(t)$ and $\delta^E(t)|_{\alpha_1=1}$ lie at the RHS of $\delta^E(t)|_{\alpha_1=0}$.

(iv) An upper bound of α_1 is provided by the observation that the necessary and sufficient condition for $\delta = \delta_c^E$ to intersect the curve $\delta^E(t)$ (see Fig.5), i.e. for (4.9) to have a real solution, is that

$$\alpha_1 \le \bar{\alpha}_1 = -1.5\,\gamma^2/\gamma^{(2)}(\delta_c - \delta_o) . \tag{4.10}$$

In all our simulations $\bar{\alpha}_1$ has been found to be larger than 0.2 , except for some few cases discussed below in (vii).

(v) Another upper bound of α_1 is provided as follows. From $\alpha_1 = 0$, Fig.5 shows that the successive $\delta^E(t)$ curves drawn for increasing values of α_1 will generally start providing a negative ϵ_2 error ($t_c^E < t_1$) , then $|\epsilon_2|$ will gradually be decreasing; in some few cases it passes by 0 , then increases afterwards (with $\epsilon_2 > 0$) . It is therefore important to upper-bound the α_1 value so as to avoid overoptimistic evaluation of the CCT :

$$\alpha_1 \le \alpha_1^* . \tag{4.11}$$

According to all our simulations α_1^* should not exceed 0.6 . In fact, we could - and do - choose two different values of α_1^* depending upon whether the fault applies at a generator bus or not; in the former case we set $\alpha_1^* = 0.3$, in the latter $\alpha_1^* = 0.6$.

(vi) Combining (4.10),(4.11), we get

$$\alpha_1 = \min[\alpha_1^*, \bar{\alpha}_1] . \tag{4.12}$$

(vii) An exception to the above considerations arises - quite seldom indeed - when γ is negative and hence $\gamma^{(2)}$ positive; according to (4.10), $\bar{\alpha}_1$ is then negative. In such cases, instead of (4.12) we just take $\alpha_1 = 0.2$.

4.4. Discussion

An approximate analytic expression of the non-linear relationship (4.1) relating ϵ , ϵ_1 , ϵ_2 is given by writing successively (see eqs.(4.2),(4.3), and Fig.3) :

$$\epsilon = (t_c^E - t_1) + (t_1 - t_c^S)$$
$$\simeq \epsilon_2 + \epsilon_1 \left[\frac{\partial\delta}{\partial t}\Big|_{t=t_c}\right]^{-1} + \frac{1}{2}\epsilon_1^2 \left[\frac{\partial^2\delta}{\partial t^2}\right]^{-2}_{t=t_c} \tag{4.13}$$

Our numerous simulations show that the contribution of the ϵ_2-type error is more important than that of the ϵ_1-type for almost all cases. For example in the case of Fig.5, we get the values ($\alpha_1^* = 0.6$) :

α_1	ϵ_1-type (s)	ϵ_2(s)	ϵ(s)
0.3	-0.050	-0.252	-0.302
$\bar{\alpha}_1 = 0.4057$		-0.118	-0.168

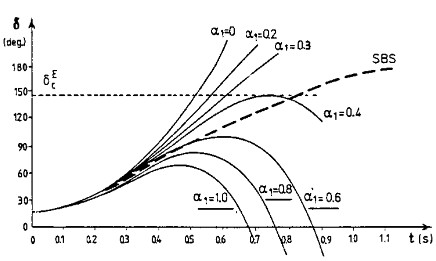


FIGURE 5

5. "SECOND SWING" INSTABILITY

5.1. Possible types of instability in the first sine-like cycle

At the beginning of the fault-on period, the OMIB may either accelerate or decelerate, as sketched in Figs. 6 and 7. If the fault is sustained long enough, this acceleration (resp. deceleration) may cause the loss of synchronism (see respectively curves a and a'). One generally refers to as the "first swing" type of instability. However, if the fault is cleared early enough, the slope of the angular evolution will generally change sign. It may then happen that the system loses synchronism because of a too large angular deviation along this new direction (see curves b and b'); we will then speak of "second swing" type of instability. If on the contrary (the disturbance having been cleared much earlier than previously), the system oscillates as in c or c', we will say that it is stable with respect to both first and second swings.

Of course, the above cases are not equally probable. First swing instability is far more frequent than second one. However, this latter must be properly identified and solved whenever it arises. Nevertheless, till now, all direct methods including the EEAC were unable to treat second swing instability. The purpose of this section is to generalize the EEAC by making it able to handle any type of instability. In what follows we will
(i) investigate the physical nature of the second swing instability;
(ii) identify all above cases by setting very simple criteria;
(iii) derive an appropriate formulation to face them.

5.2. Transient processes associated with "second swing" instability

Let us illustrate the case sketched in Fig.8, where P_m has arbitrarily been chosen negative.

Following Section 2, the largest angular deviation that the system may afford during the accelerating period without going unstable is given by

$$A(1,2,3,4) = A(4,5,6)$$

where $A(i,j,k)$ denotes the area inside the closed curve (i,j,k,i). Thus, if the fault is cleared just before δ reaches δ_{c1}, the system has enough capability of transforming kinetic to potential energy; we shall call second swing this motion during the second and third quarters of the (pseudo-) sinusoidal oscillation.

At $\delta = \delta_P$ the total potential energy that the system has accumulated during this back swing, represented by $A(1,2,3,5,8,9)$, is transformed into kinetic energy. Hence, δ will continue decreasing and the system transforming its kinetic to potential energy again, but this time its maximum capability of absorbing kinetic energy is $A(9,10,11)$. Therefore, if

$$A(9,10,11) < A(1,2,3,5,8,9)$$

the rotor angle will continue decelerating beyond $\delta = \delta_{10}$ and a second swing type of instability will arise. This corresponds to the case b of Fig.6.

To prevent the system from going "second swing" unstable, one should clear the disturbance before δ reaches δ_{c2}, which is smaller than δ_{c1}.

From a physical point of view, such a situation may arise when a disturbance applies nearby a generator and an important load bus. For example, in the basic version of the Greek system, the actual CCT of a three-phase short-

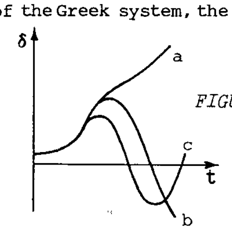


FIGURE 6

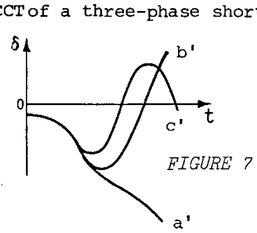


FIGURE 7

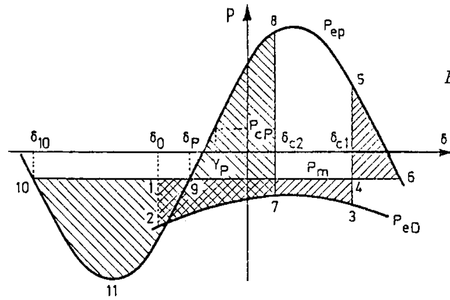


FIGURE 8

circuit applied at bus #0 is found to be 0.75 s for t^S_{c1}, 0.65 s for t^S_{c2}. By the EEAC, we compute : $t^E_{c1} = 0.747$, $t^E_{c2} = 0.654$. (Incidentally, the Liapunov criterion provides $t^D_c = 0.770$ s.)

5.3. Identification criteria

5.3.1. Second swing following an accelerated fault-on motion, see Fig.8 and curve b *of Fig.6.* Observe that according to Fig.1, the fault-on motion will be first an acceleration if

$$P_m > P_{CD} + P_{max\,D} \sin(\delta_o - \nu_D) \quad . \tag{5.1}$$

During the back swing, instability will arise if

$$A(9,10,11) < A(1,2,3,5,8,9) \quad . \tag{5.2}$$

But since

$$A(1,2,3,5,8,9) = A(9,4,5,8) + A(1,2,3,4)$$
$$\leq A(9,4,5,8) + A(4,5,6) = A(9,4,6,5,8) \;,$$

we deduce that a necessary condition of second swing instability is

$$A(9,10,11) < A(9,4,6,5,8) \tag{5.3}$$

or equivalently

$$P_m < P_{CP} \quad . \tag{5.4}$$

Notice that for a given candidate critical cluster, the condition (5.4) is fault independent and therefore extremely simple to check. (Of course, the actual CCA and CCT are fault dependent.)

5.3.2. Second swing following a decelerated fault-on motion, see Fig.9 and curve b' *of Fig.7.* The fault-on motion will be first a deceleration when inequality (5.1) is reversed :

$$P_m < P_{CD} + P_{max\,D} \sin(\delta_o - \nu_D) \quad . \tag{5.5}$$

In such a case eqs.(2.12),(2.13) are still valid, provided that the signs of P_m, P_C, ν, γ and $\gamma^{(2)}$ are inverted. From a physical viewpoint this amounts to exchanging the two equivalent machines.

During the back swing, a necessary condition of the second swing instability is given by

$$P_m > P_{CP} \quad . \tag{5.6}$$

5.4. Formulation of the second swing characteristics

The CCA of the above case 5.3.1 derives readily by transforming inequality (5.2) to an equality :

$$A(9,10,11) = A(1,2,7,8,9) \tag{5.7}$$

where

$$A(9,10,11) = 2P_{max\,P}\cos(\delta_P - \nu_P) + (P_m - P_{cP})(\pi + 2\delta_P - 2\nu_P) \tag{5.8}$$

$$A(1,2,7,8,9) = P_{max\,P}[\cos(\delta_P - \nu_P) - \cos(\delta_{c2} - \nu_P)] - (P_m - P_{cP})(\delta_{c2} - \delta_P) + P_{max\,D}[\cos(\delta_{c2} - \nu_D) - \cos(\delta_o - \nu_D)] + (P_m - P_{cD})(\delta_{c2} - \delta_o) \;. \tag{5.9}$$

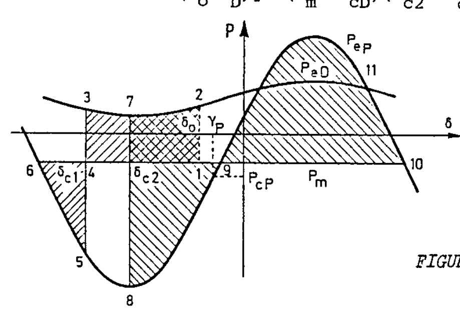


FIGURE 9

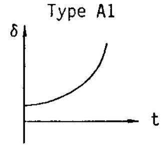
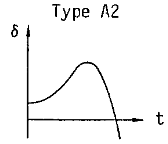
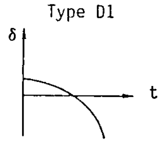
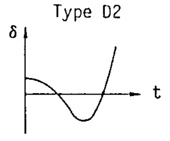
Eqs.(5.7)-(5.9) replace the first swing CCA expressions (2.12)-(2.14). The same eqs.(5.7)-(5.9) may also provide the CCA of case 5.3.2 : one merely has to invert therein the signs of P_m, P_c and ν.

In both cases, the CCT is expressed by solving(2.16).

5.5. Summary

Table 1 summarizes all possible cases of instability, along with the corresponding identification criteria and formulation.

TABLE 1

Fault-on evaluation criterion : $P_m \gtrless P_{cD} + P_{maxD} \sin(\delta_o - \nu_D)$		Type of instability: "first swing"	Type of instability: "second swing"
	Yes : forward	Type A1  criterion : $P_m > P_{cP}$ δ_c : (2.14)	Type A2  criterion : $P_m < P_{cP}$ δ_c : (5.7)-(5.9)
	No : backward	Type D1  criterion : $P_m < P_{cP}$ δ_c : (2.14) with inverted signs of P_m, P_c and ν	Type D2  criterion : $P_m > P_{cP}$ δ_c : (5.7)-(5.9) with inverted signs of P_m, P_c and ν

5.6. Remarks

1. The accuracy of the above second swing stability assessment is as good as that of the first swing.
2. The extension to the second swing assessment makes the EEAC more robust and rich in information, at the price of solely two "IF-THEN" programming statements.
3. It may happen that second swing instability depends on a critical cluster different from the first swing's one (e.g. see in § 6.5).
4. Both first swing (t_{c1}) and second swing (t_{c2}) CCTs are important for TSA. For the stability assessment stricto sensu, observe that even if t_{c2} is smaller than t_{c1}, the time to reach instability relative to the first swing type is generally much shorter than that of the second swing type. But since the two types of instability may call for opposite control actions, it is important to identify the type of instability to be controlled.
5. The second swing instability, with $t_{c2} < t_{c1}$, is certainly within the validity of the classical model. On the other hand, the second swing duration is 1 s or more, and reaches the validity limits of the simplified model. For further exploration, more sophisticated models would be needed.

6. SIMULATIONS

6.1. Purposes, conditions, and presentation

The objectives of the large body of simulations shortly described hereafter are

- to assess the EEAC robustness and accuracy for various power systems, differing by their parameters, topology, configuration, and, within a given power system, for various operating conditions;
- to identify and explore multimachine critical cluster cases;
- to identify and explore uncommonly encountered types of instability.

Ten test systems, comprising real (or reduced version of real) power systems have been considered :

- 3-machine test system;
- 6-machine Tunisian EHV simplified power system;
- 6-machine Chinese regional system;
- 7-machine CIGRE system;
- 9-machine test system;
- 10-machine New England system;
- 14-machine Greek EHV simplified power system (N.B. the configuration used here corresponds to its situation in the early seventies);
- 15-machine test system;
- 17-machine Iowa system, reduced version of the network of the State of Iowa;
- 38-machine Belgian EHV simplified power system.

To further test the EEAC robustness, we have created 26 (respectively 18) additional modified versions of the CIGRE (resp. Greek) system, by changing significantly topology, load and generator characteristics. This resulted in large changes of their operating conditions and hence of CCTs (e.g. at many generator buses, the CCT has changed as much as from 0.15 to 0.80 s.).

The disturbances are of the standard three-phase short-circuit type. All possible locations, i.e. *generator buses* (GBs) and *other buses* (OBs) are systematically scanned. Overall, this provides more than 2500 cases.

The SBS method is the benchmark for accuracy; integration steps of 1 ms. were used to properly compare accuracy provided by EEAC w.r.t. SBS.

Throughout our simulations, α_1^* has been fixed at 0.3 and 0.6 , respectively for GB and OB locations (and with $\alpha_1\alpha_2^* = 1$, as considered in Section 4). This much more stringent than necessary in practice condition, has intentionally been imposed to test the EEAC robustness. It can very easily be relaxed so as to adapt the α_1, α_2 values to the specifics of a given power system, if not of a given (important) bus of it; this exploration, performed off-line once for all, can greatly enhance the accuracy.

Statistical information is defined by means of

$$\Delta_j = (t^E_{c,j} - t^S_{c,j})/t^S_{c,j} \times 100\ \%$$

$$\mu = \sum_{j=1}^{k} \Delta_j / k \quad ; \quad \sigma = \left(\sum_{j=1}^{k} \Delta_j^2 / k \right)^{1/2}$$

where $t^E_{c,j}$, $t^S_{c,j}$ denote respectively the t^E_c and t^S_c of the j-th test, k the total number of tests.

All cases are sorted into two categories according to their t^S_c value, viz. : $t^S_c \gtrless 1$ s.

(i) If $t^S_{c,j} > 1.0$ s, we merely examine whether $t^E_{c,j} > 0.95$ s, without calculating either of the actual t^S_c and the discrepancy Δ_j ; if yes, we declare the EEAC to be satisfactory; otherwise we qualify the case as "difficult".

(ii) If $t^S_{c,j} < 1.0$ s , we calculate Δ_j, μ and σ.

Due to space limitations, the presentation of the simulation results will essentially focus on statistical gathering, and items related to Sections 3 and 5. Only the simulations of the basic case of the Greek system will be individually explicited. A much more comprehensive simulation description and analysis are reported in [5]. Computing times are also provided in [2].

6.2. Global simulation results

Table 2 gathers all explored (582 GB and 2029 OB) cases, and reports the number of large discrepancies

TABLE 2

System \ Number of cases	Total Nr. of cases: GB	Total Nr. of cases: OB	Total Nr. of cases: Total	$t^S_c < 1.0$ sec. Nr. of GB: $\lvert\Delta\rvert$ >10%	$t^S_c < 1.0$ sec. Nr. of GB: Total	$t^S_c < 1.0$ sec. Nr. of OB: $\lvert\Delta\rvert$ >10%	$t^S_c < 1.0$ sec. Nr. of OB: Total	$t^S_c > 1.0$ sec. Nr. of GB: t^E_c <0.95	$t^S_c > 1.0$ sec. Nr. of GB: Total	$t^S_c > 1.0$ sec. Nr. of OB: t^E_c <0.95	$t^S_c > 1.0$ sec. Nr. of OB: Total
Basic conf. of 10 systems	127	466	593	6	119	35	160	0	8	4	306
27 conf. of CIGRE system	189	81	270	4	185	2	29	0	4	1	52
19 conf. of GREEK system	266	1482	1748	12	245	59	233	0	21	7	1249

(i.e. of $|\Delta| > 10$ %) obtained for CCTs below and above 1 s (the latter having been qualified as "the difficult" cases); the distinction is also made between GB and OB cases as well as between basic and modified configurations. Notice that there are only 12 (all of the OB type) difficult cases; their t_c^E ranges from 0.55 to 0.95 s.

6.3. Statistical information

Fig.10 (resp. 11) shows the error distributions on all buses (respectively on the GBs only) of the basic configurations of all 10 systems. Because of fixing the same α_1^*, $\alpha_1\alpha_2^*$ values for all systems, almost all distributions have a negative bias, especially in the GB cases (see Fig.11). It is therefore suggested to set $\alpha_1\alpha_2^*$ to a value somewhat larger than 1.

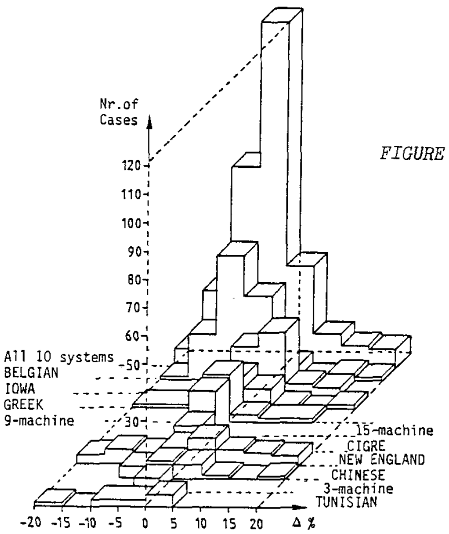


FIGURE 10

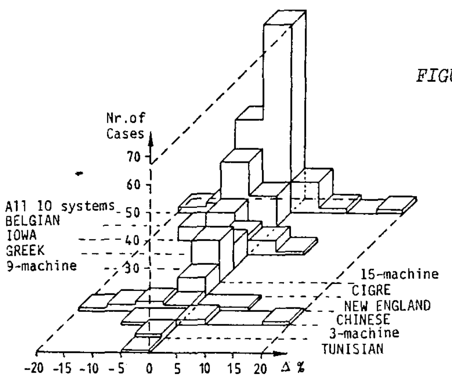


FIGURE 11

Fig.12 shows on the other hand the mean value and standard deviation of each system. The dotted line shows the $\bar{\mu}$ and $\bar{\sigma}$ of the 10 systems altogether.

Note incidentally that in the Tunisian system there is a GB with $t_c^S = 0.911$ s and $t_c^E = 0.602$ s ; this large discrepancy spoils the statistical results.

6.4. Multimachine critical cluster cases

Table 3 collects all such cases encountered during our simulations with the basic systems configurations. Except for the 2-machine clusters, the members composing each cluster are also identified.

Concerning the CIGRE system, the same 2-machine cluster prevails for most of the 27 configurations when the disturbance is located at bus #4.

The clustering of the Greek system into 4 machines (concentrated in the Northern part) and 10 machines (in the Southern part) appears in many disturbance locations with all 19 configurations. All cases with more than 2-machine critical clusters are related to this scheme. Hence, only a few candidate clusters should be added to the list provided by the accelerations criterion. As for the 2-machine critical clusters, they are automatically identified on the basis of the above list. The off-line preselection makes the on-line critical cluster identification always very little demanding.

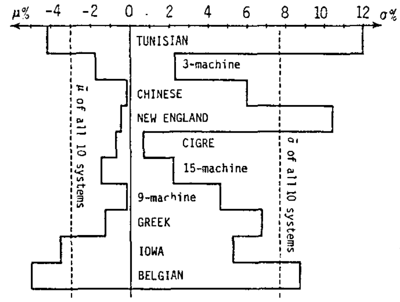


FIGURE 12

TABLE 3

	GB Number of machines	GB cases	OB Number of machines	OB cases
Tunisian			2	1
CIGRE	2	1		
9-machine			2	3
New England	2	1	2 4 (G4,5,6,7) 5 (G4,5,6,7,9)	10 1 2
GREEK	3 (North - G17) 10 (South)	1	9 (South - G59) 10 (South)	1 1
15-machine			2	1
IOWA			2 5(G2,10,12,16,17)	1 2
Belgian	2	1	2 3 (G1,17,17) 11 (G1,5,17,18,26,27,28,29,32,33,34)	11 1 1

6.5. Unusual types of instability

The type A1 instability has concerned about 95 % of our simulations. Table 4 gathers other types of instability encountered in the basic configurations (the notation is in accordance with that of Table 1).

Remarks. 1.- Types A2 and D2 are difficult to discover by the SBS method, but very easy by the EEAC. The robustness and accuracy of the latter are surprisingly

TABLE 4

type	system	GB/OB	BUS	first swing t_{c1}^S	first swing t_{c1}^E	second swing t_{c2}^S	second swing t_{c2}^E
D1	CHINESE	GB	6	.807	.993		
	NEW ENGLAND	GB	30	.484	.495		
		OB	1 3 9	.715 .254 .597	.731 .282 .717		
	IOWA	OB	25	.291	.280		
	BELGIAN	GB	MAAS1	> 1.00	1.45		
A2	3-machine	GB	ONE	.710	.704	.560	.549
	GREEK	GB	0	.750	.747	.650	.654
	IOWA	GB	G1 G4	.305 .390	.316 .359	.293 .378	.286 .345
D2	TUNISIAN	OB	OS225 NS225	.594 (A1) .532 (A1)	.591 (A1) .552 (A1)	.550 .490	.560 .497
	BELGIAN	GB	REVI1	>1.00	1.59	>1.00	1.54
		OB	AVEL1 LINT3 MAST1	>1.00 >1.00 >1.00		>1.00 >1.00 >1.00	2.36 2.42 2.92

good even for multimachine critical clusters with second swing instability.

2.- For bus #OS225 of the Tunisian system, the first swing instability is of type A1, and the critical machine is SS225, whereas the second swing instability is of type D2 and the critical machine changes into SF150.

3.- For bus #0 of the Greek system, the critical cluster contains G0, G19, G21 for both types A1 and A2.

4.- In our 18 modified versions of the Greek system, there are much more A2 cases than in the basic configuration; some of them have low t_{c2} values (< 0.35 s).

6.6. Individual CCT information for the Greek system

There are 14 GBs and 78 OBs in this system. Table 5 reports all cases corresponding to $t_c^S < 1.0$ s. Multimachine critical clusters are found for both GBs and OBs (see Table 3). The asteristic indicates that this is an A2 type instability.

TABLE 5

BUS	t_c^S (ms)	t_c^E (ms)	Δ %	Cluster S	BUS	t_c^S (ms)	t_c^E (ms)	Δ %	Cluster S
GB 1	490	533	8.78	South	GB 89	308	295	-4.22	G89
GB 33	655	633	-3.36	G33	0 t_{c1}	750	751	0.13	North (-G17)
GB 34	341	326	-4.40	G34	0 t_{c2}	650*	654*	0.61	North (-G17)
GB 64	713	687	-3.65	G64	OB 22	774	745	-3.75	G21
GB 76	483	470	-2.69	G76	OB 37	356	340	-4.49	G34
GB 17	587	596	1.53	G17	OB 71	386	368	-4.66	G34
GB 19	527	545	3.04	G19	OB 72	375	358	-4.53	G34
GB 21	506	513	1.38	G21	OB 75	473	548	15.86	G34
GB 48	>1000	982	OK	G48	OB 82	670	694	3.58	South
GB 59	488	490	0.41	G59	OB 88	900	732	-18.67	G89
GB 77	487	483	-0.82	G77	OB 91	413	431	4.36	G34
GB 84	268	261	-2.61	G84	OB 31	930	828	-10.97	South (-G59)

7. ACTUAL AND POTENTIAL APPLICATIONS

Generally speaking, the possible needs in the area of TSA may be classified in two different ways :

* according to the type of TSA : conventional analysis (in terms of CCTs and margins) vs. sensitivity analysis and therefrom control;
* according to the desired degree of modelling sophistication.

By essence, the EEAC is an extremely fast and flexible method, based on a very simplified modelling. It was initially designed for the purpose of providing either a screening tool in planning stability studies, or an on-line ultrafast technique. Let us briefly comment on its possibilities in this latter field.

Within the conventional TSA context, the EEAC has been developed in [1,2] and further generalized in this paper. The improvements and extensions brought out, contributed to make it mature enough and ready for real application. On the other hand, analytic sensitivity techniques are being developed in [3]. Among the many interesting results proposed therein, let us merely quote the straightforward calculation of generation and load supply limits, i.e. of transient stability constraints. This opens real possibilities towards on-line transient security control.

Besides its use as such, the EEAC may also serve as an auxiliary technique within more refined methods, using sophisticated system modelling. For example, within the artificial intelligence-based approach proposed in [6,7], it has provided a systematic means to build learning sets and to preclassify their operating points.

The above are only samples of the EEAC possibilities.

8. CONCLUSION

Three main objectives have been pursued in this paper. They all aimed at enhancing the EEAC's accuracy and robustness. They have been reached at the expense of negligible additional computing effort which, according to the conclusions of [2] is much lower than that of the most efficient direct methods.

Computational efficiency [2] together with good accuracy make the EEAC mature enough, i.e. ready for real world applications.

A sample of domain applications where the EEAC proves to be very useful have been briefly scanned.

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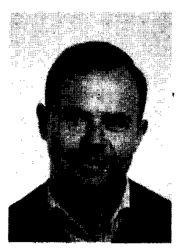
Yusheng Xue was born in Jiangsu, China, in 1941. He received the BSSE from Shandong Institute of Technology in 1963 and the MSSE from EPRI, China in 1981. He is currently nearing completion of his Ph.D. degree from the University of Liège, Belgium, on leave from NARI, China, where he is the depute director of the System Engineering Department and in charge of power system application software.



Thierry Van Cutsem was born in Charleroi, Belgium, in 1956. He graduated in electrical engineering from the University of Liège, Belgium, in 1979. He received the Agrégé de l'enseignement supérieur degree (Ph.D. degree) from the same University in 1984. Since 1979, he has been a research assistant in the Department of Electrical Engineering. His main interest is power system analysis and control, with special emphasis on state estimation and transient stability.

Mania Ribbens-Pavella was born in Nafplion, Greece. She received the electrical (electronics) engineering degree and the degree of Docteur es Sciences Appliquées in 1958 and 1969 respectively from the University of Liège, where she is currently Professor in the Department of Electrical Engineering. Her research interests lie in the field of electric power system analysis and control.

Discussion

Sebastião C. M. de Oliveira (Centro de Pesquisas de Energia Elétrica–CEPEL, Rio de Janeiro, Brazil): This is a very interesting paper which makes a real contribution to the subject of transient stability assessment of multimachine power systems. The authors have to be congratulated by the contribution for a better understanding of how second swing justability develops. The easy detection of this kind of instability for the one machine-infinite bus system is shown.

The idea of system reduction in two and only two equivalent clusters is also very interesting because it allows to reduce the difficulties of the transient stability assessment for multimachine systems to the analysis of an equivalent one machine-infinite bus system. Accordingly, we must consider, beyond the stable equilibrium point δ_p, only one unstable equilibrium point $(\Pi - \delta_p + 2v_p)$.

While the authors claim that the system reduction criterion for two equivalent clusters is more sound that the conventional one and state that the extended equal area criterion is able to handle any type of instability, it is timely to remember the A1, A2, D1 and D2 types of instability presented and theoretically characterized are referred to the case of one machine-infinite bus system. Those types of instability cannot comprise all instability situations verified in multimachine power systems.

In other words, the external system to a certain generation plane (or group of plants) which instability is being evaluated may interact with this plant through more than one dominant electromechanical oscillation frequency. In this case, therefore, reduction of the global system to only two equivalent cluster cannot lead, in general, to a consistent and accurated evaluation of the stability limits by using the extended equal area criterion. The appropriateness of these comments is confirmed by the observations of item 3.1 with regard to deterioration of the extended equal area criterion for clearing critical time evaluation in case of multimachine critical clustering. For disturbances located at a center of a group of plants or nearby weak interconnections, several electromechanical modes of oscillations are generally excited with non neglegible amplitudes. Comments of the authors with respect to these observations are appreciated.

On the other hand, very often, only one electromechanical oscillation frequency is adequate to characterize the external system dynamics. This explains the consistent results of critical clearing time evaluation of most instability cases reported in the paper.

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Y. Xue, Th. Van Cutsem and **M. Ribbens-Pavella**: We thank Dr. de Oliviera for his interest in our paper. We indeed believe that the notion of partial center of angles (PCOA) is sounder than that of the conventional (global) center of angles. Our simulation results consistently corroborate this conjecture. This concept is used for detecting A_1, A_2, D_1 and D_2 types of instability; the fact that these types of instability are referred to the case of one-machine-infinite-bus system simply comes from the replacement of a two-machine equivalent, precisely constructed on the basis of two PCOAs.

We agree, that these types of instability cannot comprise all instability situations verified in multimachine power systems. But this fact does not infirm the results provided by EEAC: this criterion aims at reflecting the stability behaviour of the system *as a whole* and at detecting the first departure of one critical cluster with respect to the others. Admittedly, EEAC is not intended to describe internal, sometimes quite intricate, electromechanical oscillation modes.

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